

WASP-10b

R-1887

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_130721 · created 2026-08-03T12:37:52+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456494.9305 +/- 0.002 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.21 +/- 0.061
Transit depth (Rp/Rs)^2	4.42 +/- 2.57 [%]
Orbital Inclination (inc)	89.65 +/- 1.91
Airmass coefficient 1 (a1)	0.119 +/- 0.087
Airmass coefficient 2 (a2)	1.86 +/- 1.54
Scatter in the residuals of the lightcurve fit is	59.47 %
Best Comparison Star	#3 - [290, 345]
Optimal Aperture	1.81
Optimal Annulus	6.76
Transit Duration (day)	0.0943 +/- 0.0094

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.21 ± 0.061 vs 0.1592 (deviation 0.83σ)
Duration fit vs published	0.0943 d vs 0.0946 d (deviation 0.03σ)
Mid-time O–C	-3.9 min
Depth SNR	3.4
Residual scatter	59.47 %

Light curve

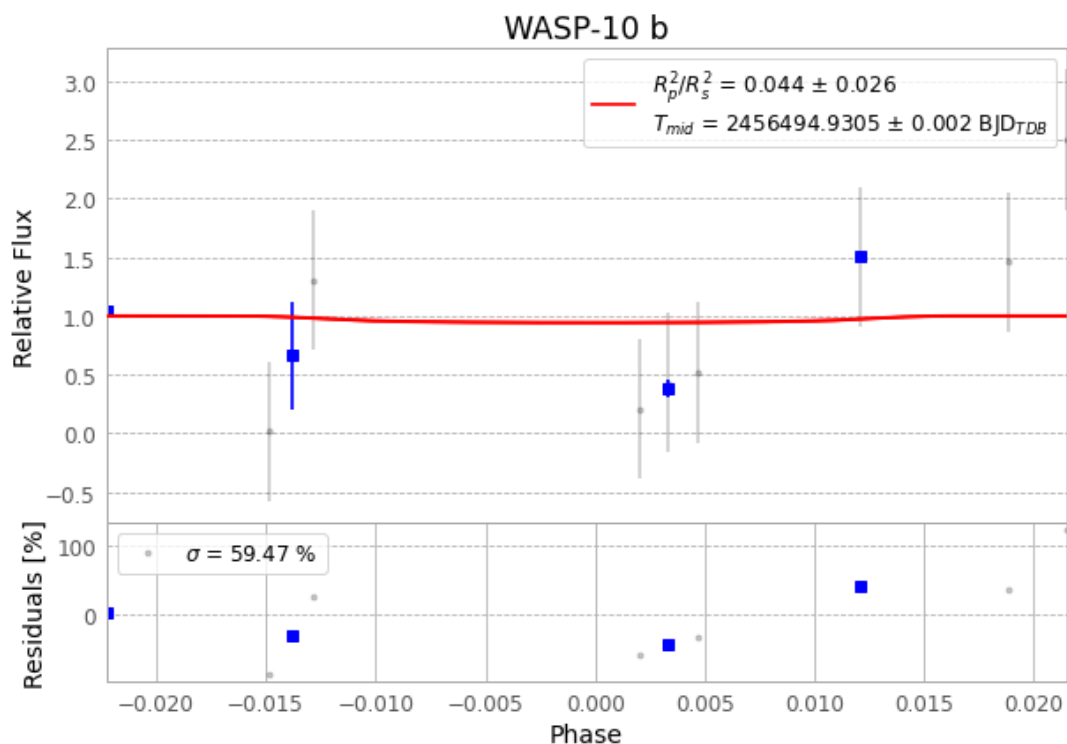


Figure 1 — FinalLightCurve_WASP-10 b_2013-07-21.png (SHA-256 433753856747a3b9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [312, 191], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a7bdf903c62b3b84...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

65 FITS frames (43 MB), WASP-10130721083313.FITS → WASP-10130721115705.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-130721.log (026789c871aa...), FinalLightCurve_WASP-10 b_2013-07-21.png (433753856747...), FinalLightCurve_WASP-10 b_2013-07-21.csv (ed23bc181bdd...)

Compute resources

Wall time 485.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 12:37Z.